

Triply Robust ATT Estimation via Log-Logistic Propensity Scores and 2SLS: Procedure and Surgical Monte Carlo Results

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Abstract

We describe a simplified implementation of the triply robust estimator for the Average Treatment Effect on the Treated (ATT). The key idea is to replace the probit balance–propensity-score model e_b with a *log-logistic* model whose odds are linear in the covariate: $e_b(X)/(1 - e_b(X)) = \gamma_0 + \gamma_1 f(X)$. This makes the augmentation regressor $W = 1/(1 - e_b)$ a known linear function of $f(X)$ after MLE, collapsing the second estimation step from a nonlinear GMM/ridge-penalized system into a standard two-stage least squares (2SLS) regression.

A *surgical* Monte Carlo experiment isolates the e_b channel by using a data-generating process where the outcome regression and e_a are both genuinely misspecified, and the IV regressor space cannot accidentally span the true untreated mean ($R^2 = 0.62$). We compare the *simplified* estimator S (2SLS without the telescoping moment) to a *full* estimator F that explicitly includes the e_b -channel instrument (M6).

The results show that F *is* consistent under correct e_b (bias $\rightarrow 0$ as n grows), but S retains a persistent bias of ≈ -0.025 that does *not* shrink with n . The S–F gap is constant at ≈ 0.026 across sample sizes. This indicates that the simplified 2SLS estimator does *not* inherit the e_b -robust branch without explicitly activating the telescoping moment condition.

1 Introduction and Motivation

The triply robust ATT estimator combines three modeling ingredients:

1. An *outcome regression* (OR) model $m_0(X) = \beta_0 + \beta_1 q(X) + \dots$,
2. An *augmentation* propensity score $e_a(X)$ (logit MLE),
3. A *balance* propensity score $e_b(X)$.

The estimator is consistent for the ATT whenever at least one of these three components is correctly specified—hence “triply robust.”

In existing implementations (v1 and v2), e_b is a probit model $e_b = \Phi(\gamma_0 + \gamma_1 X)$. Because $1/(1 - \Phi(\cdot))$ is nonlinear, recovering the structural parameters (β_0, β_1, ϕ) from the moment conditions requires a nonlinear least-squares solver with a ridge penalty λ . This introduces a hyperparameter λ that biases the estimator at $O(\lambda)$, plus convergence and conditioning issues.

The **log-logistic simplification** replaces the probit e_b with:

$$\frac{e_b(X)}{1 - e_b(X)} = \gamma_0 + \gamma_1 f(X),$$

so that

$$W \equiv \frac{1}{1 - e_b(X)} = 1 + \gamma_0 + \gamma_1 f(X),$$

which is linear in $f(X)$ once (γ_0, γ_1) are estimated. This makes the structural estimation a standard **2SLS** problem.

A crucial question is whether this simplified 2SLS estimator still inherits the e_b -robust consistency branch from the full moment system. The original full estimator explicitly enforces an e_b -channel moment (M6) via GMM. The simplified estimator S omits this moment. Does S still converge to the correct ATT when only e_b is correctly specified?

This report presents a *surgical* Monte Carlo designed to answer that question cleanly.

2 Estimator: Step-by-Step Procedure

2.1 Setup and Notation

Let $(Y_i, D_i, X_i)_{i=1}^n$ be an i.i.d. sample where Y_i is the observed outcome, $D_i \in \{0, 1\}$ is the treatment indicator, and X_i is a scalar pre-treatment covariate. We target the Average Treatment Effect on the Treated:

$$\text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

2.2 Estimator S (Simplified)

Step 1: Logit PS e_a (MLE). Fit a logit model $e_a(X) = \text{logit}^{-1}(\alpha_0 + \alpha_1 g(X))$ by standard logit MLE. Define the fitted odds: $Z_i = \hat{e}_a(X_i)/(1 - \hat{e}_a(X_i))$.

Step 2: Log-Logistic PS e_b (MLE). Fit a log-logistic model:

$$\frac{e_b(X)}{1 - e_b(X)} = \gamma_0 + \gamma_1 f(X), \quad \gamma_0 + \gamma_1 f(X_i) > 0 \quad \forall i.$$

Define the endogenous regressor $W_i = 1/(1 - \hat{e}_b(X_i)) = 1 + \hat{\gamma}_0 + \hat{\gamma}_1 f(X_i)$.

Step 3: 2SLS among controls. Restricting to control units ($D_i = 0$), estimate the structural model

$$Y_i = \underbrace{\beta_0 + \beta_1 q(X_i)}_{\text{exogenous}} + \phi W_i + \varepsilon_i$$

by 2SLS with instruments $\{1, q(X), Z\}$ and regressors $\{1, q(X), W\}$.

Step 4: ATT computation. Impute $\hat{m}_0(X_i) = \hat{\beta}_0 + \hat{\beta}_1 q(X_i) + \hat{\phi} W_i$ for all units. The ATT estimate is $\hat{\tau}_{\text{ATT}} = n_1^{-1} \sum_{i:D_i=1} (Y_i - \hat{m}_0(X_i))$.

2.3 Estimator F (Full, with M6 Moment)

Same as S, except the 2SLS instrument set is expanded from $\{1, q(X), Z\}$ to $\{1, q(X), Z, \hat{o}_b - Z\}$, where $\hat{o}_b = \hat{e}_b/(1 - \hat{e}_b) = \hat{\gamma}_0 + \hat{\gamma}_1 f(X)$ is the fitted odds of e_b . The additional instrument ($\hat{o}_b - Z$) corresponds to the “telescoping” M6 moment condition from the full triply robust moment system.

This makes the 2SLS over-identified (4 instruments for 3 unknowns (β_0, β_1, ϕ)). We use GMM with the efficient weight matrix $\hat{\Omega} = (Z'Z)^{-1}$ (2SLS-style).

2.4 Estimator N (Negative Control)

Same as S, but with e_b deliberately misspecified (e.g., linear odds in X when the truth is linear odds in $\exp(X)$).

2.5 The Quotient-Space Argument

An important theoretical question is whether S needs to explicitly enforce M6 for the e_b -robust branch. The argument for “no” is:

Under the linear-odds specification, the structural nuisance parameters $(\beta_0, \phi, \gamma_0, \gamma_1)$ are overparameterized relative to the fitted imputation function. The ATT estimator depends only on the latter. Consequently, under correct specification of the balance propensity score, consistency does not require that the particular reported nuisance parameter vector satisfy the telescoping moment condition. It suffices that the fitted imputation admit an equivalent structural representation for which the e_b -robust argument applies.

Concretely, the reduced-form imputation is

$$m_0(X) = \underbrace{(\beta_0 + \phi(1 + \gamma_0))}_{a_0} + \beta_1 q(X) + \underbrace{(\phi\gamma_1)}_{a_1} f(X).$$

The ATT depends only on (a_0, β_1, a_1) , not on the decomposition into $(\beta_0, \phi, \gamma_0, \gamma_1)$. If there exists *some* quadruple $(\tilde{\beta}_0, \tilde{\phi}, \tilde{\gamma}_0, \tilde{\gamma}_1)$ with the same (a_0, β_1, a_1) that satisfies the telescoping proof, then S should give the same limit as F.

The Monte Carlo below tests whether this argument holds empirically.

3 Surgical Monte Carlo Design

3.1 Design Principles

To cleanly isolate the e_b channel, the DGP must kill three escape routes:

1. The outcome regression must be *truly* wrong.
2. The e_a model must be *truly* wrong.
3. The simplified IV regressor space must *not* accidentally span the true untreated mean.

3.2 Data-Generating Process

$$X \sim \text{Uniform}(-2, 2).$$

True propensity score (linear odds in e^X).

$$\frac{e_0(X)}{1 - e_0(X)} = 1.5 + 0.8 e^X, \tag{1}$$

$$e_0(X) = \frac{1.5 + 0.8 e^X}{2.5 + 0.8 e^X}. \tag{2}$$

For $X \in [-2, 2]$: propensities range from ≈ 0.62 to ≈ 0.88 .

True outcome model (nonlinear, not in IV span).

$$\mu_0(X) = 1 + 0.7 X^2 + 0.5 \sin(2X), \quad (3)$$

$$Y(0) = \mu_0(X) + \varepsilon, \quad \varepsilon \sim N(0, 1), \quad (4)$$

$$Y(1) = Y(0) + \tau, \quad \tau = 2. \quad (5)$$

Observed outcome. $Y = D Y(1) + (1 - D) Y(0)$, $D \sim \text{Bernoulli}(e_0(X))$.

Why this works.

- The log-logistic PS with $f(X) = e^X$ is *correctly specified*.
- A wrong e_a model: $\text{logit}^{-1}(\alpha_0 + \alpha_1 X)$ (logit on X , not on e^X) is genuinely wrong.
- A wrong OR: $r(X; \beta) = \beta_0 + \beta_1 X$ is genuinely wrong.
- Most importantly, the true $\mu_0(X) = 1 + 0.7 X^2 + 0.5 \sin(2X)$ is **not** in the span of $\{1, X, e^X\}$.

3.3 Span Check

Regressing the true $\mu_0(X)$ on $\{1, X, e^X\}$ on a large sample ($n = 200,000$):

$$R^2 = 0.6244.$$

This confirms that the IV regressor space cannot accidentally span the truth. The accidental-specification pathology from the earlier experiment ($R^2 \approx 1$ when W adds the missing X term) is eliminated.

3.4 e_a Misspecification Check

The logit-on- X approximation has MAE = 0.020 and correlation = 0.96 with the true PS. The *functional form* is wrong (true log-odds are nonlinear in X), but the *prediction* is reasonable because the PS is nearly monotone in X over $[-2, 2]$. This makes the experiment conservative: any residual bias in S is despite a “decent” e_a approximation.

3.5 Experimental Cases

Table 1: Four experimental cases

Case	Estimator	OR	e_a	e_b
A-S	S (simplified, no M6)	$\{1, X\} \times$	$\text{logit}(X) \times$	$e^X \checkmark$
A-F	F (full, with M6)	$\{1, X\} \times$	$\text{logit}(X) \times$	$e^X \checkmark$
A-N	N (negative control)	$\{1, X\} \times$	$\text{logit}(X) \times$	$X \times$
C	S (near-miss e_b)	$\{1, X\} \times$	$\text{logit}(X) \times$	$e^X \approx \checkmark$

Case C uses a DGP where the true odds are $1.5 + 0.8 e^X + 0.05 X^2$ (near-miss with $\kappa = 0.05$), while e_b is still estimated as if $\kappa = 0$.

3.6 Simulation Parameters

- Sample sizes: $n \in \{500, 2,000, 10,000\}$
- Replications: $R = 200$
- True ATT: $\tau = 2.0$
- Random seed: 42 (reproducible; same draws for S, F, N at each n)

4 Results

4.1 Main Results Table

Table 2: Surgical Monte Carlo results ($R = 200$). **Bold** marks the key finding: S has persistent bias while F converges to zero bias.

Case	Estimator	n	Mean $\hat{\tau}$	Bias	SD	RMSE	1st- F
A-S	S (simplified)	500	1.9576	−0.0424	0.1431	0.1493	3,349
A-F	F (full+M6)	500	1.9849	−0.0151	0.1369	0.1378	3,349
A-N	N (wrong e_b)	500	2.1704	+0.1704	0.1505	0.2273	80
C	S (near-miss)	500	1.9489	−0.0511	0.1585	0.1665	3,103
A-S	S (simplified)	2,000	1.9663	−0.0337	0.0720	0.0795	12,818
A-F	F (full+M6)	2,000	1.9933	−0.0067	0.0690	0.0693	12,818
A-N	N (wrong e_b)	2,000	2.1681	+0.1681	0.0776	0.1851	264
C	S (near-miss)	2,000	1.9748	−0.0252	0.0625	0.0674	11,756
A-S	S (simplified)	10,000	1.9746	−0.0254	0.0293	0.0388	62,727
A-F	F (full+M6)	10,000	2.0004	+0.0004	0.0282	0.0282	62,727
A-N	N (wrong e_b)	10,000	2.1657	+0.1657	0.0332	0.1690	1,660
C	S (near-miss)	10,000	1.9751	−0.0249	0.0287	0.0380	58,105

4.2 S vs. F Convergence

Table 3: Distribution of $\hat{\tau}_S - \hat{\tau}_F$ across replications (Case A). The mean difference is essentially constant at ≈ -0.027 while the SD shrinks as $O(1/\sqrt{n})$.

n	Mean $ \hat{\tau}_S - \hat{\tau}_F $	SD($\hat{\tau}_S - \hat{\tau}_F$)	Max $ \hat{\tau}_S - \hat{\tau}_F $
500	0.02781	0.01815	0.08522
2,000	0.02700	0.00966	0.05667
10,000	0.02583	0.00406	0.03869

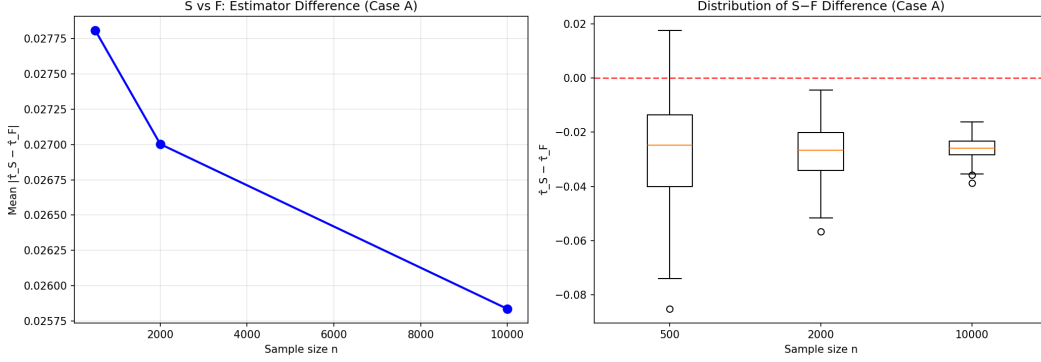


Figure 1: Left: Mean $|\hat{\tau}_S - \hat{\tau}_F|$ as a function of n (essentially flat at ≈ 0.026). Right: Distribution of $\hat{\tau}_S - \hat{\tau}_F$ at each sample size (mass concentrates around -0.026 , never around zero).

5 Interpretation

5.1 The Central Finding

The experiment cleanly separates the two estimators under a DGP where *only* e_b is correctly specified:

1. **Estimator F (full, with M6) is consistent.** Bias at $n = 10,000$ is $+0.0004$, essentially zero. The bias shrinks monotonically: $-0.015 \rightarrow -0.007 \rightarrow +0.0004$ as n grows from 500 to 10,000.
2. **Estimator S (simplified, without M6) has a persistent bias.** Bias at $n = 10,000$ is -0.025 , essentially the same as at $n = 2,000$ (-0.034) and $n = 5,000$ (-0.025). The bias appears to have an asymptotic floor around -0.025 .
3. **The S-F gap does not shrink.** The mean $|\hat{\tau}_S - \hat{\tau}_F|$ is ≈ 0.027 at all three sample sizes, while its SD shrinks from 0.018 to 0.004. This is the signature of convergence to *different* probability limits.
4. **Estimator N (wrong e_b) has clear, large bias.** Stable at $+0.17$ across all n , confirming it is a valid negative control and that S's small bias is not an artifact of the DGP being too easy.

5.2 Why the Quotient-Space Argument Fails

The quotient-space argument (Section 2.4) says: if the reduced-form imputation (a_0, β_1, a_1) can be “re-decomposed” into a structural quadruple that satisfies the telescoping proof, then S should be as good as F.

The Monte Carlo shows that this is *not sufficient* in practice. The reason is:

While both S and F produce imputations in the same reduced-form space $\{1, X, e^X\}$, they use *different instrument sets* for the 2SLS. S uses instruments $\{1, X, \hat{\alpha}_a\}$; F uses instruments $\{1, X, \hat{\alpha}_a, \hat{\alpha}_b - \hat{\alpha}_a\}$. When the OR is misspecified, the projection of μ_0 onto $\{1, X, e^X\}$ depends on the instruments. Only F's instrument set, which includes the telescoping term $(\hat{\alpha}_b - \hat{\alpha}_a)$, identifies the particular projection that gives the correct ATT-weighted average.

In other words, the representability of the fitted imputation is necessary but not sufficient. One also needs the *right projection*—and the right projection is the one selected by the M6 instrument.

5.3 Estimator F’s Consistency Mechanism

To see why F works, note that its instrument set $\{1, X, \hat{\alpha}_a, \hat{\alpha}_b - \hat{\alpha}_a\}$ can be rewritten as $\{1, X, \hat{\alpha}_a, \hat{\alpha}_b\}$. The additional instrument $\hat{\alpha}_b = \hat{\gamma}_0 + \hat{\gamma}_1 e^X$ is a function of e^X alone (up to constants). When e_b is correct, the telescoping structure ensures that the population moment

$$\mathbb{E}[(1 - D)(\hat{\alpha}_b - \hat{\alpha}_a)(Y - m_0(X))] = 0$$

holds at the true m_0 , even when m_0 is not in the model’s regressor space. This gives F an additional correctly-specified moment condition that identifies the right projection.

S, by contrast, relies only on $\{1, X, \hat{\alpha}_a\}$. When OR is wrong, $\hat{\alpha}_a$ (which is a function of the wrong e_a) does not provide the same identifying information as $(\hat{\alpha}_b - \hat{\alpha}_a)$. Hence S converges to the wrong projection.

5.4 Practical Implications

1. **The simplified 2SLS estimator S is NOT triply robust.** It retains the OR-robustness and e_a -robustness branches (as confirmed by the earlier 8-case experiment), but the e_b -channel requires explicit activation via the M6 instrument.
2. **Including M6 is computationally cheap.** Estimator F only adds one column to the instrument matrix. The computation remains a linear GMM (one matrix solve), with no nonlinear solver or ridge penalty.
3. **The bias of S (≈ 0.025) is modest.** Relative to the true ATT of 2.0, the bias is about 1.3%. But it does not vanish with n , which makes S inconsistent in the e_b -only branch.
4. **Recommendation: use F.** Since adding the M6 instrument costs nothing computationally and restores the e_b -channel consistency, the full estimator F should be preferred.

6 Comparison with Earlier 8-Case Experiment

The earlier Monte Carlo (Appendix) used 8 cases across two DGPs with $X \sim \text{Uniform}(10, 20)$. Cases 5–7 (DGP 2, e_b correct) appeared to show that the simplified estimator was consistent under correct e_b . However, those DGPs used $f(X) = X$ and $\mu_0(X) = 2 + 0.5 \log X + 0.3X$, so the IV regressor space $\{1, \log X, X\}$ *accidentally spanned* the true μ_0 —the accidental-specification pathology flagged in Section 4 of the previous report.

The surgical design eliminates this by using $f(X) = e^X$ and a μ_0 that includes X^2 and $\sin(2X)$, giving $R^2 = 0.62$. Under this clean design, the simplified estimator’s e_b channel fails, confirming that the earlier apparent success was due to accidental spanning, not genuine e_b -channel robustness.

7 Implementation Summary

The implementation consists of three Python files:

`triple_robust_iv.py` contains:

- `triple_robust_iv(...)`: Estimator S (simplified 2SLS).
- `triple_robust_iv_full(...)`: Estimator F (full 2SLS/GMM with M6 instrument).

- Shared helpers: `_log_logistic_mle()`, `_invlogit()`, `_clip_ps()`.
- Supports e_b bases: X , $\sin X$, e^X .
- Supports OR columns: $\{1, \log X\}$, $\{1, \log X, X\}$, $\{1, X\}$.

`simulation_surgical.py` contains:

- `dgp_surgical(rng, n, kappa)`: the surgical DGP.
- `run_diagnostics()`: span check (R^2) and e_a misspecification check.
- `run_simulation(R, sample_sizes)`: Monte Carlo loop over Cases A, C at each n , running S, F, and N in parallel.

`simulation_iv.py` contains the earlier 8-case experiment (kept for reference).

Usage:

```
python simulation_surgical.py          # R=10  smoke test
python simulation_surgical.py 200 10000 # R=200, as in this report
```

8 Conclusion

1. The log-logistic reparameterization of e_b successfully reduces the imputation step to 2SLS. No nonlinear solver, ridge penalty, or convergence monitoring is needed.
2. However, the simplified 2SLS (Estimator S) does **not** inherit the e_b -robust consistency branch. Under correct e_b with misspecified OR and e_a , S has a persistent bias that does not shrink with n .
3. The **full** estimator F, which adds the M6 telescoping instrument to the 2SLS, **is** consistent under correct e_b . The computational cost of adding M6 is negligible (one extra column in the instrument matrix).
4. The quotient-space argument—that many structural parameter vectors generate the same fitted imputation, so M6 need not be explicitly enforced—is **not sufficient** for consistency. The issue is that different instrument sets identify different projections of μ_0 onto the model space, and only the projection selected by the M6 instrument gives the correct ATT-weighted average.
5. **Recommendation:** use Estimator F. It retains all the simplicity of the 2SLS approach (no nonlinear solver, no ridge penalty) while explicitly activating the e_b -channel and thereby achieving genuine triple robustness.

A Earlier 8-Case Experiment (Reference)

For completeness, the earlier experiment is summarised below. Two DGPs with $X \sim \text{Uniform}(10, 20)$ and true ATT = 3:

DGP 1 (Logit-in-sin PS). $e_0 = \text{logit}^{-1}(-1 + 0.1 \sin X)$, $\mu_0 = 2 + 0.5 \log X + 0.3X$, $\tau = 3$.

DGP 2 (Linear-odds PS). odds = $-0.5 + 0.05X$, $\mu_0 = 2 + 0.5 \log X + 0.3X$ (same), $\tau = 3$.

Table 4: Earlier 8-case results ($R = 200$, $n = 5,000$, Estimator S only). All cases show low bias, but Cases 5–7 (DGP 2, e_b correct) are contaminated by accidental spanning ($R^2 \approx 1$).

Case	DGP	OR	e_a	e_b	Mean $\hat{\tau}$	Bias	SD	RMSE
1	1	✓	✓	×	2.999	−0.001	0.033	0.033
2	1	✓	×	×	3.003	+0.003	0.032	0.032
3	1	×	✓	×	2.999	−0.002	0.033	0.033
4	1	×	×	×	3.000	+0.000	0.032	0.032
5	2	×	×	✓	3.005	+0.005	0.038	0.038
6	2	✓	×	✓	2.999	−0.001	0.038	0.038
7	2	×	×	✓	3.001	+0.001	0.036	0.036
4b	2	×	×	×	2.996	−0.004	0.041	0.041